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B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024
Second Semester

21ECC101J - ELECTRONIC SYSTEM AND PCB DESIGN

(For the candidates admitted from the academic year 2021-2022 to 2023-2024)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer ALL Questions

Marks BL CO PO

- When a pentavalent impurity is added to a pure semiconductor, it becomes _____.
A) An Insulator
B) An Intrinsic Semiconductor
C) P-type Semiconductor
D) N-type Semiconductor
- An Intrinsic semiconductor at absolute zero temperature behaves as _____.
A) an insulator
B) a metallic conductor
C) a superconductor
D) a semiconductor
- In an n-type semiconductor the intrinsic carrier concentration and hole concentration are given as, $n_i = 1.5 \times 10^{15} \text{ cm}^{-3}$ and $n = 2.25 \times 10^8 \text{ cm}^{-3}$, respectively. Then the electron concentration is in the order of
A) 10^{38}
B) 10^{22}
C) 10^{23}
D) 10^7
- The Common Base (CB) amplifier compared to CE and CC amplifier _____.
A) Higher input resistance
B) Larger voltage gain
C) Larger current gain
D) Lower input resistance
- When a reverse bias voltage exceeding the breakdown voltage is applied to an IMPATT diode, it results in
A) Breakdown
B) Saturation current
C) Avalanche multiplication
D) depletion region
- Which of the following is NOT a property of a Schottky diode?
A) Lower forward voltage drop
B) Higher switching speed
C) Lower Knee or Corner voltage
D) Lower forward current
- In order to turn on the SCR we need to provide appropriate _____.
A) Break over Voltage
B) Anode current
C) Gate Current
D) Cathode Current
- Which of the following devices has a peak point, a valley point and a negative resistance region in its IV-characteristics?
A) Schottky diode
B) Gunn diode
C) PNP diode
D) IMPATT diode

9. The ripple factor and rectification efficiency of a half wave rectifier is _____. 1 2 3 1
 A) 1.21 and 0.812 B) 0.48 and 0.406
 C) 1.21 and 0.406 D) 0.48 and 0.812
10. A CRO is an instrument designed to graphically present amplitude as a function of _____. 1 2 3 0
 A) frequency B) phase angle
 C) time D) frequency & Phase
11. Pick the odd application of thyristors. 1 1 3 1
 A) Switch for power control B) Power regulator
 C) Audio Amplifier D) Power drive controls
12. Which is not the correct statement with respect to SMPS. 1 1 3 1
 A) Higher efficiency and Low cost B) Roughed and Low ripple factor
 C) Stable output and Low ripple factor D) Less complex circuit and Low ripple factor
13. _____ printed circuit boards have wiring patterns on both sides of the insulating material. 1 1 4 1
 A) single sided B) Double side
 C) Multilayer D) rigid and flexible
14. _____ file is used to generate the artwork on a computer controlled plot. 1 1 4 1
 A) CAD B) PAM
 C) ADC D) DAC
15. Application of the photoresist is carried out in a machine called a _____. 1 1 4 1
 A) Photolithographer B) Laminator
 C) Photo printer D) Epitaxy
16. The spacing between voltage and ground conductors should be _____ as possible to avoid cross-talk problems. 1 1 4 1
 A) small B) Large
 C) Near D) Long
17. When designing a layout, it is a good practice to provide for a _____ area for any modification which may be required on the board. 1 2 5 2
 A) 10 to 20 % B) 5 to 10 %
 C) 20 to 30 % D) 50 to 70%
18. For _____ components, the leads are bent approximately 90° and inserted into the holes on the PCB and soldered. 1 2 5 2
 A) Axial B) Radial
 C) SMT D) Electromechanical
19. In common consideration all analog circuit PCB designs is to keep the signal conductor as _____ as possible. 1 1 5 1
 A) long B) Short
 C) Medium D) too short
20. ANSI abbreviated as _____. 1 1 5 1
 A) American National Standards Institute B) American Nature Standards Institute
 C) Academic National Standards Institute D) Artificial National Standards Institute

PART - B (5 x 8 = 40 Marks)**Marks BL CO PO**

Answer ALL Questions

- 21 a. In a P-type germanium, $n_i = 2.1 \times 10^{19} \text{ m}^{-3}$ and density of Boron is $4.5 \times 10^{23} \text{ atoms/m}^3$. The electron and hole mobilities are $0.4 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$ and $0.2 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$, respectively. Estimate the conductivity before and after addition of Boron atoms.

(OR)

- b. Derive the Einstein relationship for semiconductors.
- 22 a. With a neat sketch, Explain the construction, and working principle of a thyristor device.
- b. With relevant diagrams explain the steps involved in the Monolithic IC fabrication.
- 23 a. Draw the circuit diagram and explain the working of Monostable Multivibrator.

(OR)

- b. A Full wave rectifier has a load of 100Ω . If the diode resistance and secondary coil resistance together have resistance of 120Ω and the input voltage has a signal voltage of peak value 12 V then calculate (i) Peak, average and rms value of current flowing (ii) dc power output and, (iii) efficiency.
- 24 a. Explain in detail about Leaded Through-hole Assembly and Surface Mount Assembly.
- b. Classify printed circuit boards based on usage and applications, layers and also mention the advantages.
- 25 a. Briefly explain about the thermal management in the design of PCBs.
- b. Write about packaging density on PCB.

PART - C (1 x 15 = 15 Marks)**Marks BL CO PO**

Answer ANY ONE Question

26. Draw and explain the Common Emitter (CE) configuration and describe the input & output characteristics. Also define active, saturation and cut-off regions.
27. Sketch the flowchart and describe the manufacturing process of double side PCB.
